

Manan Kapoor

AI / ML Engineer

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EXPERIENCE

Jun 2026 – Jul 2026

Gurugram, India

LLM Evaluation · Red-Teaming · Agent Memory · Prompt Caching · Full-Stack QA

AI Engineering Intern — Colab91

- Red-teamed LLM agents with **50+ adversarial probes**, surfacing and documenting failure modes across evaluation runs.
- Diagnosed **agent memory and context-drift** issues in multi-turn interactions, driving memory design and prompt-caching improvements.
- Owned **full-stack QA** across the LLM pipeline, from evaluation to end-to-end checks.

Jan 2025 – Present

Patiala, India

NLP · Instruction Tuning · Dataset Engineering · Unsloth · HuggingFace

Research Intern — Thapar Institute of Engineering & Technology

- Built **PRISM**, the first multi-task Punjabi instruction dataset for controlled text rewriting — **16.43M tokens** from 600+ curated sources, with a Gemini 2.0 Flash teacher pipeline generating **91,000+** instruction-conditioned samples across 6 tasks.
- Full fine-tuned **LLaMA 3.1 8B** on A100 GPUs with Unsloth, producing consistent Gurmukhi output across all 6 tasks where the base model fails.
- Benchmarked the fine-tuned model against **four baseline models** across five evaluation metrics.

PROJECTS

RAG · Hybrid Retrieval · Agents

RAG Support Triage Agent

- Support-triage agent over a **117-document corpus** using hybrid **dense + BM25** retrieval with hard grounding — **zero external lookups**.
- Enforced safety via forbidden-pattern filtering, post-generation guardrails, and deterministic escalation across a **16-module pipeline**; evaluated on **29 real support tickets**.

github.com/manankapoor23/rag-support-triage-agent

LangChain · LLMs · SQL · Slack

Natural Language to SQL Query System

- Slack data assistant that turns plain-English questions into **schema-aware SQL**, executes it against a live database, and returns structured results in-channel.

github.com/manankapoor23/NL-SQL

TECHNICAL SKILLS

Languages

Python, C++, JavaScript, SQL, R

ML & NLP

LLM fine-tuning (full & QLoRA), instruction tuning, RAG, transformers, benchmarking, prompt engineering

Frameworks

PyTorch, HuggingFace Transformers, Unsloth, LangChain, OpenAI SDK, sentence-transformers, scikit-learn, Pandas, NumPy

Backend & Data

FastAPI, Docker, model serving, inference pipelines, PostgreSQL, FAISS, Chroma, Git/GitHub, Linux

EDUCATION

2024 – 2028

Punjab, India

Bachelor of Engineering, Computer Engineering

Thapar Institute of Engineering & Technology · CGPA 8.19 / 10.00